

Lab 08: Z-Transform Analysis

```
clc;
clear;
close all;

%%Task 1: Symbolic Z-Transforms
syms n z

x1_n = heaviside(n);
x2_n = (0.5)^n * heaviside(n); % 0.5^n u[n]
x3_n = dirac(n); % delta[n]

% Convertign to z domain using symbolic toolbox
X1_z = ztrans(x1_n, n, z);
X2_z = ztrans(x2_n, n, z);
X3_z = ztrans(x3_n, n, z);
check_x1 = iztrans(X1_z, z, n);

fprintf('Task 1 Results:\n');
```

Task 1 Results:

```
disp(X1_z); disp(X2_z); disp(X3_z);
```

$$\frac{1}{z-1} + \frac{1}{2}$$

$$\frac{1}{2z-1} + \frac{1}{2}$$

$$ztrans(\delta(n), n, z)$$

```
%% Task 2: Eigenfunction Property

n_vec = 0:40;
omega = pi/4;
x_input = exp(1j * omega * n_vec);

% System coefficients from: y[n] - 0.5y[n-1] = x[n]
num = 1;
den = [1, -0.5];

% Run the signal through the system
y_output = filter(num, den, x_input);

% Analytical check
z_val = exp(1j * omega);
H_val = 1 / (1 - 0.5 * z_val^-1);

% This is the theoretical output
```

```
y_theory = H_val * x_input;
```

```
% Display H in rectangular form
```

```
fprintf('Task 2: Calculated H(ej*pi/4) in rectangular form:\n');
```

Task 2: Calculated $H(e^{j\pi/4})$ in rectangular form:

```
disp(H_val);
```

```
1.1907 - 0.6512i
```

```
% Visual Verification
```

```
figure(1);
```

```
subplot(2,1,1);
```

```
plot(n_vec, real(y_output), 'b', n_vec, real(y_theory), 'r--');
```

```
title('Task 2: Real Part Comparison');
```

```
legend('Measured', 'Theory');
```

```
grid on;
```

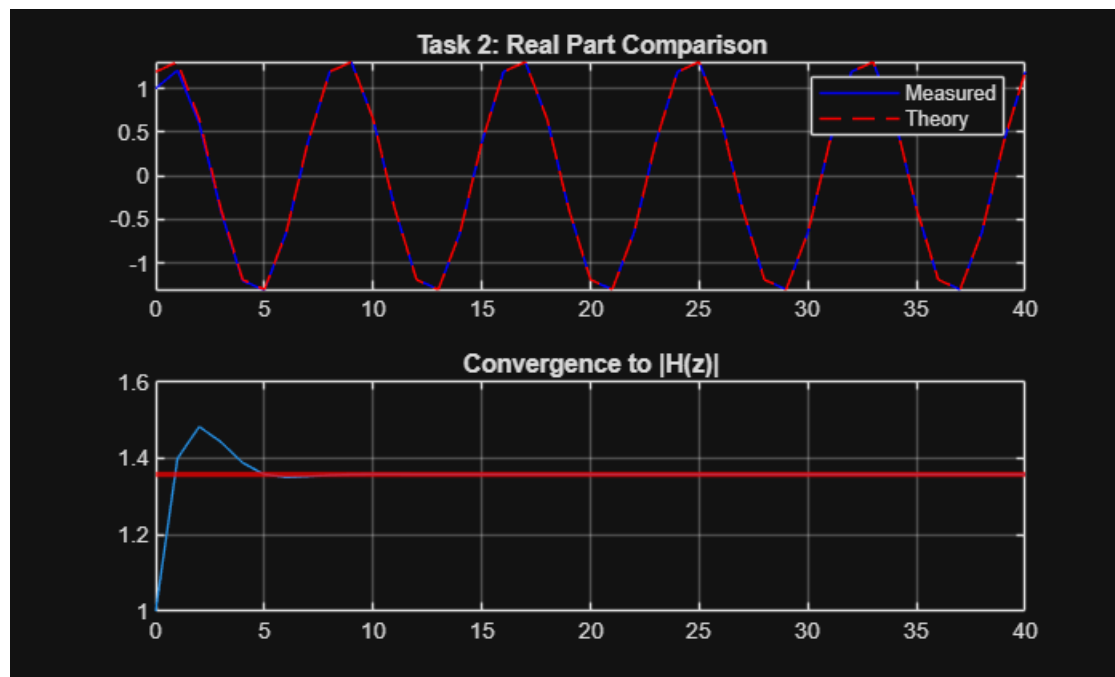
```
subplot(2,1,2);
```

```
plot(n_vec, abs(y_output ./ x_input));
```

```
ylines(abs(H_val), 'r', 'LineWidth', 2);
```

```
title('Convergence to H(z)');
```

```
grid on;
```



```
%% Task 3: Loop vs Z-Transform
```

```
% Manually solving the difference equation
```

```
N_samples = 10;
```

```
y_loop = zeros(1, N_samples);
```

```
impulse = [1, zeros(1, N_samples-1)]; %  $x[n] = \delta[n]$ 
```

```

% Recursive loop
for k = 1:N_samples
    if k == 1
        y_loop(k) = impulse(k);
    else
        y_loop(k) = impulse(k) + 0.8 * y_loop(k-1);
    end
end

% Analytical check using inverse Z-transform
%  $H(z) = 1 / (1 - 0.8z^{-1}) \rightarrow h[n] = (0.8)^n u[n]$ 
n_idx = 0:N_samples-1;
y_analytical = (0.8).^n_idx;

% format result into a table
ResultsTable = table(n_idx', y_loop', y_analytical', ...
    'VariableNames', {'n', 'Loop Method', 'Z-Transform Method'});
disp('Task 3: Comparison Table');

```

Task 3: Comparison Table

```
disp(ResultsTable);
```

n	Loop Method	Z-Transform Method
0	1	1
1	0.8	0.8
2	0.64	0.64
3	0.512	0.512
4	0.4096	0.4096
5	0.32768	0.32768
6	0.26214	0.26214
7	0.20972	0.20972
8	0.16777	0.16777
9	0.13422	0.13422

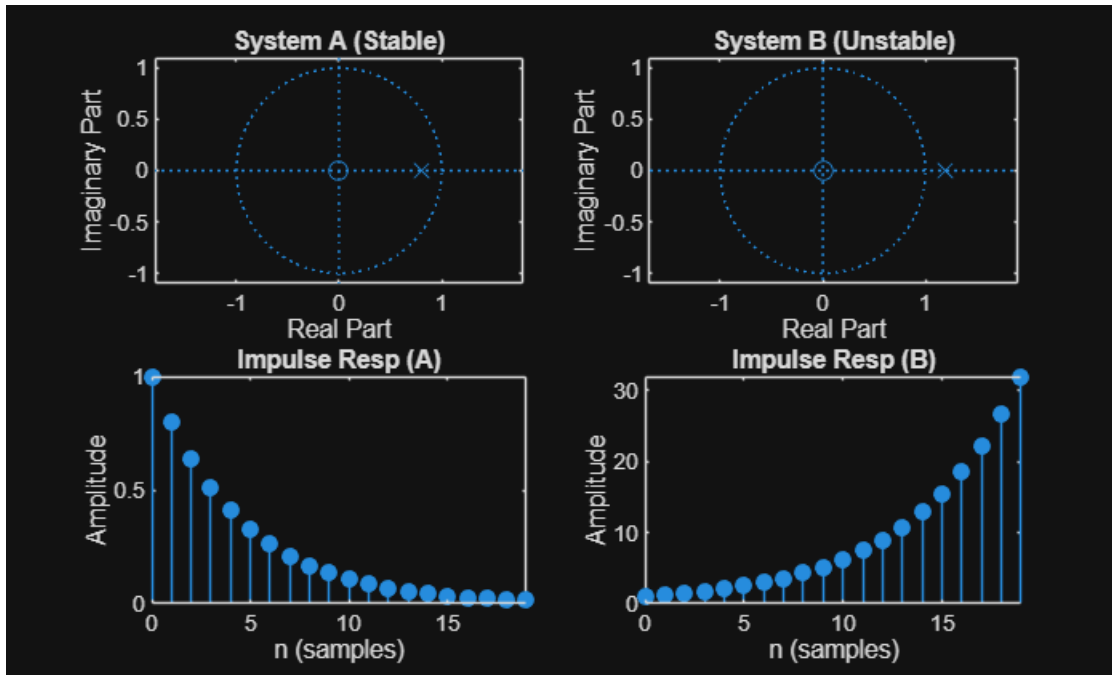
```

%% Task 4: Stability & Poles
% System A pole at 0.8
% System B pole at 1.2

figure(2);
subplot(2,2,1); zplane(1, [1, -0.8]); title('System A (Stable)');
subplot(2,2,2); zplane(1, [1, -1.2]); title('System B (Unstable)');

% Look at the Impulse Responses
subplot(2,2,3); impz(1, [1, -0.8], 20); title('Impulse Resp (A)');
subplot(2,2,4); impz(1, [1, -1.2], 20); title('Impulse Resp (B)');

```



```
% System B is unstable because its pole  $|1.2| > 1$ 
```

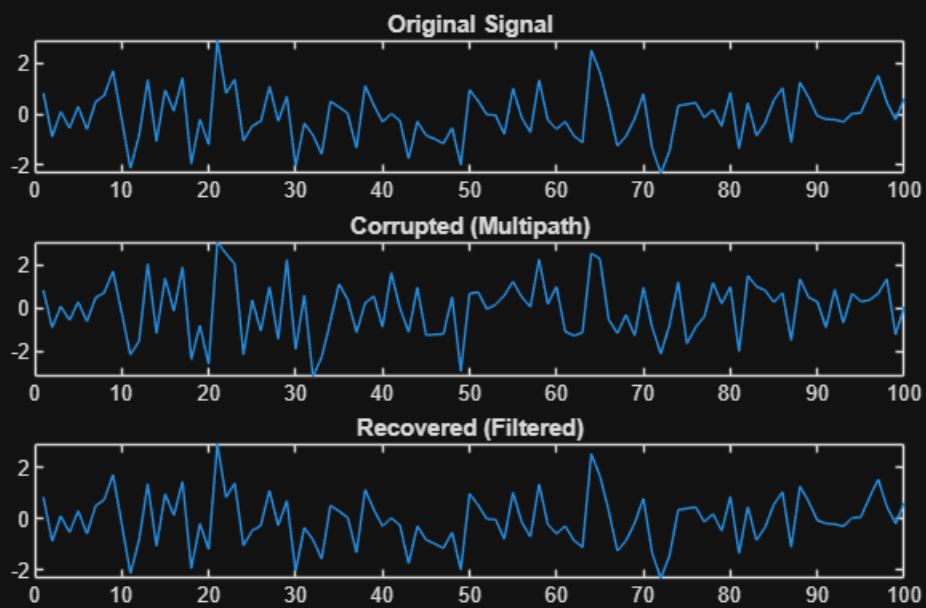
System A is stable because its pole is at 0.8, which is inside the unit circle. If you give this system a quick nudge with an impulse, the output eventually settles back to zero. It stays within bounds.

System B is unstable because its pole is at 1.2, which sits outside the unit circle. Because the pole is greater than 1, the system has a feedback effect that grows over time instead of decaying. If you look at the step response or impulse response plots, you will see the values climbing higher and higher toward infinity rather than leveling off.

```
%% Task 5: Wireless Multipath Recovery
% alpha = 0.8;
delay = 11;
original_x = randn(1, 100); % Random data

% Distortion:  $y[n] = x[n] - \alpha * x[n - \text{delay}]$ 
b_distort = [1, zeros(1, delay-1), -alpha];
corrupted_y = filter(b_distort, 1, original_x);
% We put the distortion coefficients in denominator to flip effects
recovered_x = filter(1, b_distort, corrupted_y);

figure(3);
subplot(3,1,1); plot(original_x); title('Original Signal');
subplot(3,1,2); plot(corrupted_y); title('Corrupted (Multipath)');
subplot(3,1,3); plot(recovered_x); title('Recovered (Filtered)');
```



Assessment Rubric
Lab 08
Lab 8: Z-Transform

Name:	Student ID:
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Marks distribution:

	Tasks	LR2 Code	LR5 Results	AR4 Report Submission	AR7 Comments
In-Lab	Task 1	/05	/05	/5	/5
	Task 2	/10	/10		
	Task 3	/10	/10		
	Task 4	/10	/10		
	Task 5	/10	/10		
Total =100		/45	/45	/5	/5
CLO Mapped		CLO1	CLO1	CLO3	CLO3
CLO		Total Points		Points Obtained	
1		80			
3		20			
Total		100			

For description of different levels of the mapped rubrics, please refer the provided Lab Evaluation Assessment Rubrics and Affective Domain Assessment Rubrics.